AdventureWorks Sales Data Analysis Report

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1. Executive Summary

This report analyzes the performance of AdventureWorks Cycles for the period 2015–2017, using sales, customer, product, and returns data. Data was cleaned and integrated using SQL, then analyzed with Python (Pandas, Matplotlib, Seaborn) to derive actionable insights.

Key Findings:

· Premium products (24.8% of transactions) contribute 92.3% of total profit, despite their lower sales volume.

· Sales show consistent annual growth, with a clear peak in December (holiday season).

· High-selling accessories (helmets, tires) have the highest return rates, but financial losses are larger for premium products due to their high prices.

· Outliers were detected in Profit (22.32%) and in Revenue/Price/Cost (24.85%); they represent premium products and were retained for separate analysis.

· A multiple linear regression model explains 100% of profit variance, confirming a purely linear relationship.

Recommendations:

1. Focus marketing and inventory on premium products, especially before December.

2. Improve quality of high-volume accessories to reduce return rates.

3. Consider price adjustments for regular products to boost revenue while monitoring demand elasticity.

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2. Introduction

2.1 Problem Background

AdventureWorks Cycles seeks to understand its financial drivers and identify improvement opportunities. This analysis examines:

· Profitability by product segment (regular vs. premium).

· Seasonal sales patterns.

· Return causes and financial impact.

· Factors influencing profit.

2.2 Objectives

1. Clean and integrate multi-source data.

2. Quantify profit contribution of premium products.

3. Analyze temporal sales trends.

4. Identify top-returned products and associated losses.

5. Build a regression model to explain profit drivers.

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3. Data

3.1 Source

CSV files from the AdventureWorks sample database were used:

Table	Rows	Description
sales	56,046	Order details (date, product, customer, quantity)
customers	18,000	Customer demographics (income, education)
products	~600	Product information (price, cost, color)
returns	1,809	Return records (product, territory, date)

3.2 Data Cleaning

Performed using SQL and Python:

· Duplicates removed – no duplicates found in key tables.

· Missing values – ProductColor (50 missing) filled with 'unknown'.

· Text standardization – all text columns converted to lowercase and trimmed.

· Date conversion – OrderDate converted to datetime; Year and Month extracted.

· Outlier handling – outliers in Profit (22.32%), Revenue, Price, and Cost (24.85%) were identified; they represent premium products and were kept, creating two segments for further analysis.

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4. Exploratory Analysis

4.1 Descriptive Statistics

Table 1: Summary of key numerical columns

Column	Mean	Std	Min	Max
Revenue	444.5	835.9	2.29	3,578.27
Profit	186.58	343.6	1.43	1,487.84
OrderQuantity	1.5	0.6	1	3
ProductPrice	438.96	838.65	2.29	3,578.27

4.2 Outliers & Product Segmentation

Outliers were present in 24.85% of transactions for Revenue, ProductPrice, and ProductCost, and in 22.32% for Profit. These represent premium products. Data was split into two segments:

Segment	Rows	Avg Profit	Total Profit	Contribution %
Normal	42,117	17.37	731,411	6.99%
Premium	13,929	698.27	9,726,169	93.01%

Insight: Premium products (24.8% of rows) generate 92.3% of total profit.

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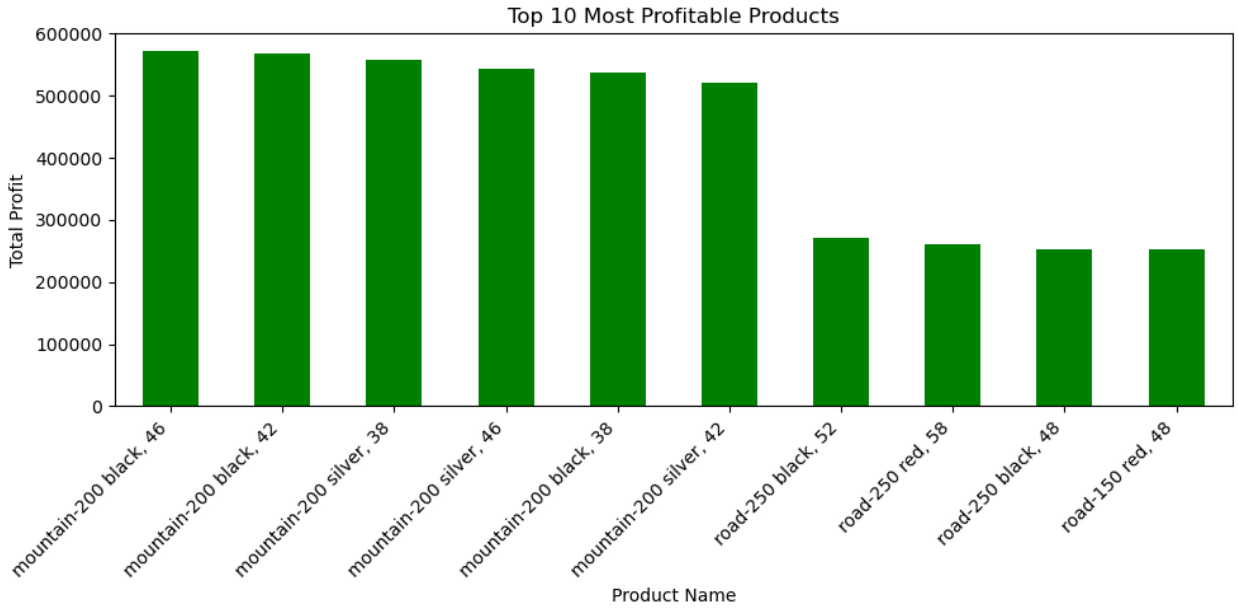
5. Product Analysis

5.1 Most Profitable Products

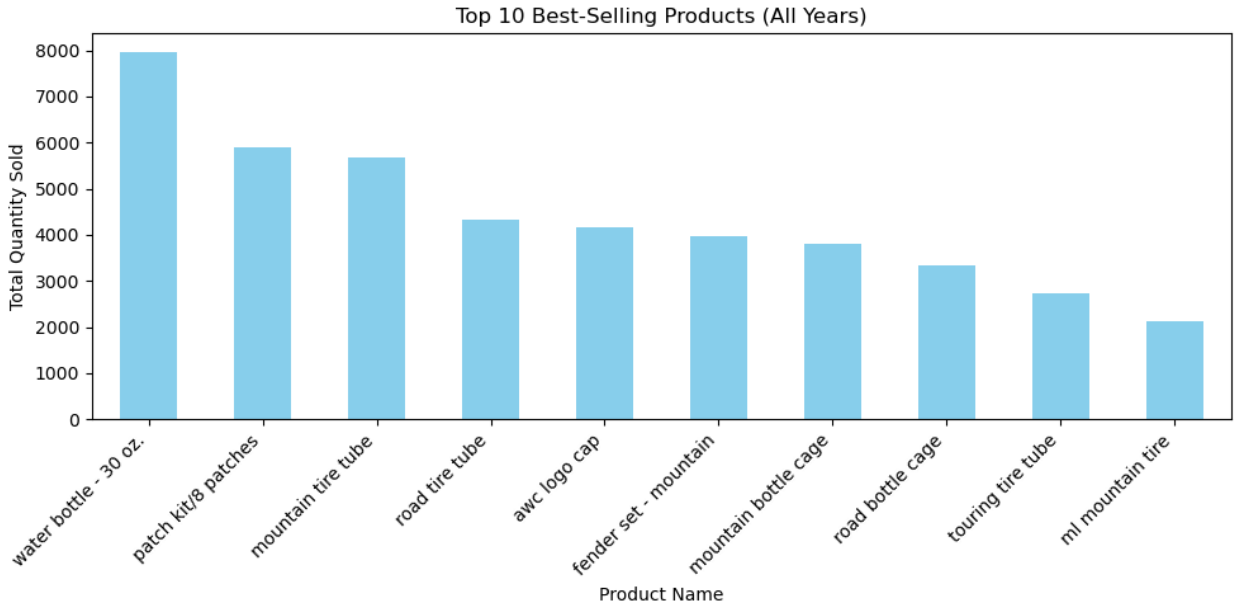
Figure 1: Top 10 most profitable products



Premium mountain bikes (mountain-200 series) dominate the list, each generating over $500k in profit.

5.2 Best-Selling Products (by quantity)

Figure 2: Top 10 best-selling products



Accessories (water bottles, tires, helmets) sell in high volumes but contribute little to overall profit.

5.3 Profit Margin

· Regular products: 57.5% (high efficiency, low price).

· Premium products: 41.1% (lower efficiency, high absolute profit).

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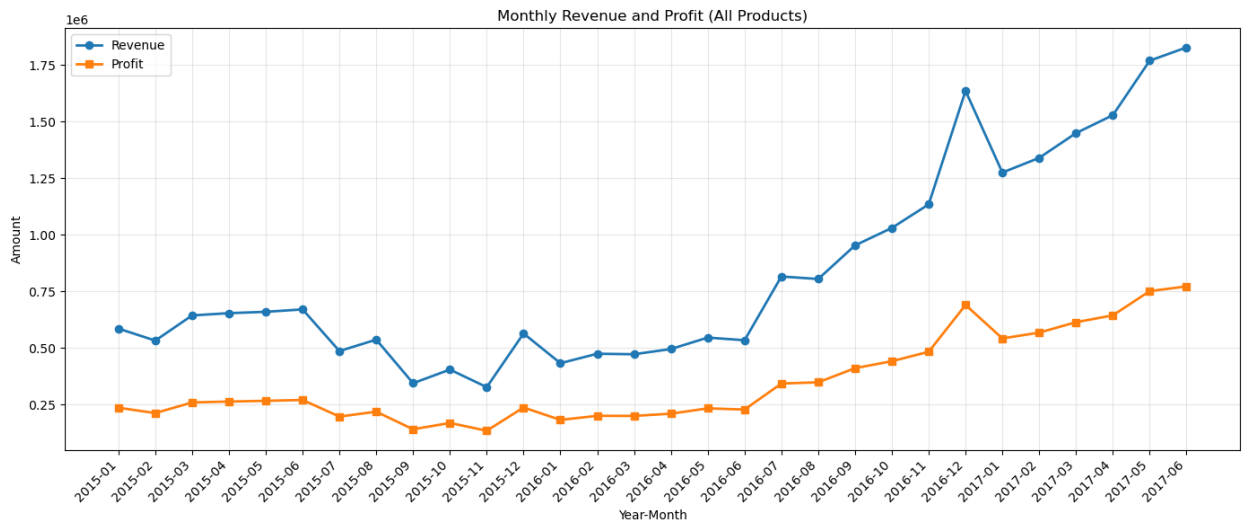
6. Temporal Analysis

6.1 Yearly Performance

Figure 3: Monthly revenue and profit (2015–2017)



· Steady year-over-year growth (2015 < 2016 < 2017).

· December peaks every year – strong holiday season effect.

Table 2: Annual totals

Year	Total Profit	Total Revenue	Orders
2015	2.6M	6.4M	2630
2016	3.9M	9.3M	10695
2017	3.8M	9.1M	11839

Note:

Data for 2017 is only available for the first six month. Therefore, total revenue and profit for 2017 are lower than for 2016.

6.2 December Analysis

In December, premium products account for 92% of profit even though they represent only 16% of units sold. Regular products sell more units but contribute only 8% of profit.

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7. Returns Analysis

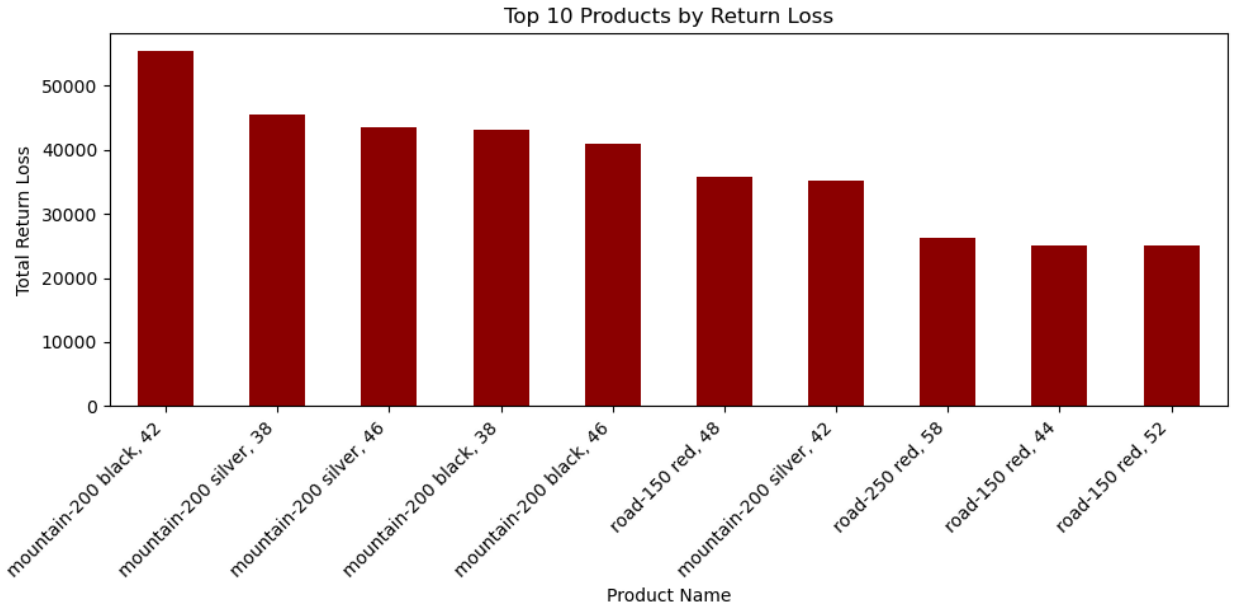
7.1 Top Returned Products (by quantity)

Figure 4: Top 10 products by return quantity

Accessories (water bottles, tires, helmets) have the highest return volumes due to their high sales volume.

7.2 Top Returned Products (by financial loss)

Figure 5: Top 10 products by return loss



Despite low return rates, premium products cause the largest financial losses when returned because of their high prices.

7.3 Profit vs. Return Rate

Product	Total Profit	Return Rate %
mountain-200 black, 46	571633.74	3.30
mountain-200 black, 42	567860.58	4.49
mountain-200 silver, 38	558786.16	3.75
mountain-200 silver,46	544482.76	3.68
mountain-200 black, 38	536732.01	3.69
mountain-200 silver, 42	521597.32	3.11
road-250 black, 52	272038.08	3.48
road-250 red, 58	260846.64	3.96
road-250 black, 48	253098.72	3.40
road-150 red, 48	251849.42	5.59
road-150 red, 62	237779.62	3.55
road-150 red, 52	236372.64	4.17
road-250 black, 58	230715.84	2.24
road-250 black, 44	226411.44	3.80
road-150 red, 56	220895.86	3.82
road-150 red, 44	195570.22	5.04
touring-1000 blue, 46	158774.88	4.55
touring-1000 yellow, 46	151557.84	2.38
road-250 red, 48	148854.16	3.11
road-350-w yellow, 40	148435.20	3.33

Premium products combine high profit with low return rates, making them highly attractive.

8. Statistical Modeling

8.1 Multiple Linear Regression

A regression model was built to predict Profit using:

· OrderQuantity

· ProductPrice

· AnnualIncome

· ProductCost

Results:

· $R^2 = 1.000$ (perfect fit – profit is directly computed from these variables).

· All predictors are statistically significant ($p < 0.05$).

· The relationship is purely linear, confirming that no non-linear interactions are needed.

Supplementary Analysis: Geographic Distribution of Revenue (Using Power BI)

Following the primary analysis conducted in Python, an additional analysis was performed using Power BI to explore the **geographic distribution of revenue**. This step aimed to understand the company's performance across different markets and identify the regions generating the highest revenue.

Key Findings

Australia region recorded the highest revenue, significantly outperforming other areas, indicating strong purchasing power and effective market penetration.

Regions such as **Southwest** and **Northwest** showed solid performance, though lower than their **Australia** counterparts.

Central, Southeast and **Northeast** registered the lowest revenue, suggesting untapped growth opportunities or local challenges that require further investigation.

Recommendations

Strengthen resource allocation—particularly inventory of premium products—in top-performing region (**Australia**) especially during peak seasons.

Conduct further research to understand the underperformance in regions like **Central, Southeast** and **Northeast** , and evaluate opportunities to improve marketing or distribution strategies.

Consider targeting new markets with characteristics similar to successful regions, based on these findings.

Note: This analysis was performed in Power BI as a complement to the primary Python analysis.

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9. Conclusions & Recommendations

9.1 Conclusions

1. Premium products are the primary profit engine (92% of profit from 24.8% of transactions).

2. Sales grow annually, with a strong December peak driven by premium products.

3. High-volume accessories have the highest return rates, but their financial impact is smaller than that of premium returns.

4. Profit is perfectly explained by quantity, price, and cost – a linear relationship.

9.2 Recommendations

9.2.1 Inventory & Marketing

· Increase inventory of premium products well before December to capture holiday demand. Even though their unit sales are lower, their profit contribution justifies higher stock levels.

· Targeted marketing campaigns for premium bikes during Q4.

9.2.2 Quality & Returns

· Investigate quality issues for frequently returned accessories (helmets, tires) – even small improvements can reduce return volume and handling costs.

· Monitor return rates for premium products; while currently low, any increase would be costly.

9.2.3 Pricing Strategy

· Regular products have a high profit margin (57%) but low prices. A modest price increase could boost revenue without heavily affecting demand, provided price elasticity is carefully studied.

· Premium products already have high absolute profit; consider bundling with accessories to further increase basket value.

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10. Appendices

Appendix A: Key Code Snippets

Full code is available on GitHub. Below are essential steps used in the analysis.

```
# Data import and cleaning
import pandas as pd
import mysql.connector

conn = mysql.connector.connect(host='localhost', user='root', password='***',
database='adventureworks')
df = pd.read_sql("SELECT * FROM sales_master", conn)
conn.close()

# Date processing
df['OrderDate'] = pd.to_datetime(df['OrderDate'])
df['Year'] = df['OrderDate'].dt.year
df['Month'] = df['OrderDate'].dt.month

# Outlier detection (IQR method)
Q1 = df['Profit'].quantile(0.25)
Q3 = df['Profit'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers = df[(df['Profit'] < lower) | (df['Profit'] > upper)]
```

Appendix B: Variable Definitions

Variable	Description
Revenue	OrderQuantity × ProductPrice
Profit	OrderQuantity × (ProductPrice – ProductCost)
OrderQuantity	Number of units sold
ProductPrice	Selling price per unit
ProductCost	Manufacturing cost per unit
ReturnQuantity	Units returned
ReturnLoss	ReturnQuantity × ProductPrice